Deep Hybrid CNN-BiLSTM Network with Attention for Multi-Class Arrhythmia Diagnosis

Applied Machine Learning (AML)

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Abstract

Cardiovascular diseases, particularly cardiac arrhythmias, require timely and accurate diagnosis to prevent mortality. This report details the development of an automated Deep Learning system designed to classify five specific cardiac conditions: Sinus Bradycardia (SB), Sinus Rhythm (SR), Atrial Flutter (AFL), Sinus Tachycardia (ST), and Atrial Fibrillation (AFIB). Utilizing a large-scale dataset of 45,152 recordings, we implement a **Hybrid Architecture** that synergizes **1D Convolutional Neural Networks (CNNs)** for local feature extraction with **Bidirectional LSTMs** for temporal sequencing. Building upon foundational CNN-LSTM frameworks, this model enhances the spatial extraction phase with **Residual Blocks (ResNet)** and **Squeeze-and-Excitation (SE)** modules. The model achieves an overall accuracy of 91%, with an AUC of 1.00 for sinus anomalies.

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1 Introduction

Electrocardiogram (ECG) interpretation is a complex task requiring the identification of subtle morphological changes (P-wave presence, QRS width) and rhythmic irregularities (R-R interval consistency). While Deep Learning has shown promise, standard Convolutional Neural Networks (CNNs) often struggle to capture the long-term temporal dependencies inherent in conditions like Atrial Fibrillation.

This project implements a **Hybrid Spatiotemporal Network**, a methodology inspired by the work of Eleyan and Alboghbaish [1], who demonstrated the efficacy of combining CNNs with LSTMs for ECG analysis. By extending their approach, we replace standard convolutional layers with a deep **1D-ResNet** backbone enhanced by **Squeeze-and-Excitation (SE)** blocks. This architecture extracts robust morphological features from raw signals before passing them to a **Bidirectional LSTM (BiLSTM)** to analyze heart rhythm dynamics.

1.1 Target Class Definitions

The model is trained to detect the following SNOMED-CT encoded conditions:

- **Sinus Bradycardia (SB):** Regular rhythm, rate < 60 bpm.
- **Sinus Rhythm (SR):** Normal cardiac activity (60–100 bpm).
- **Sinus Tachycardia (ST):** Regular rhythm, rate > 100 bpm.
- **Atrial Flutter (AFL):** Sawtooth P-waves, rapid atrial rate.
- **Atrial Fibrillation (AFIB):** Irregularly irregular rhythm, absent P-waves.

2 Data Pipeline & Preprocessing

The dataset utilized is the *Large Scale 12-Lead Electrocardiogram Database for Arrhythmia Study* (PhysioNet) [2], containing 45,152 patient records sampled at 500Hz.

2.1 Data Flow Diagram

Figure 1 illustrates the end-to-end processing pipeline.

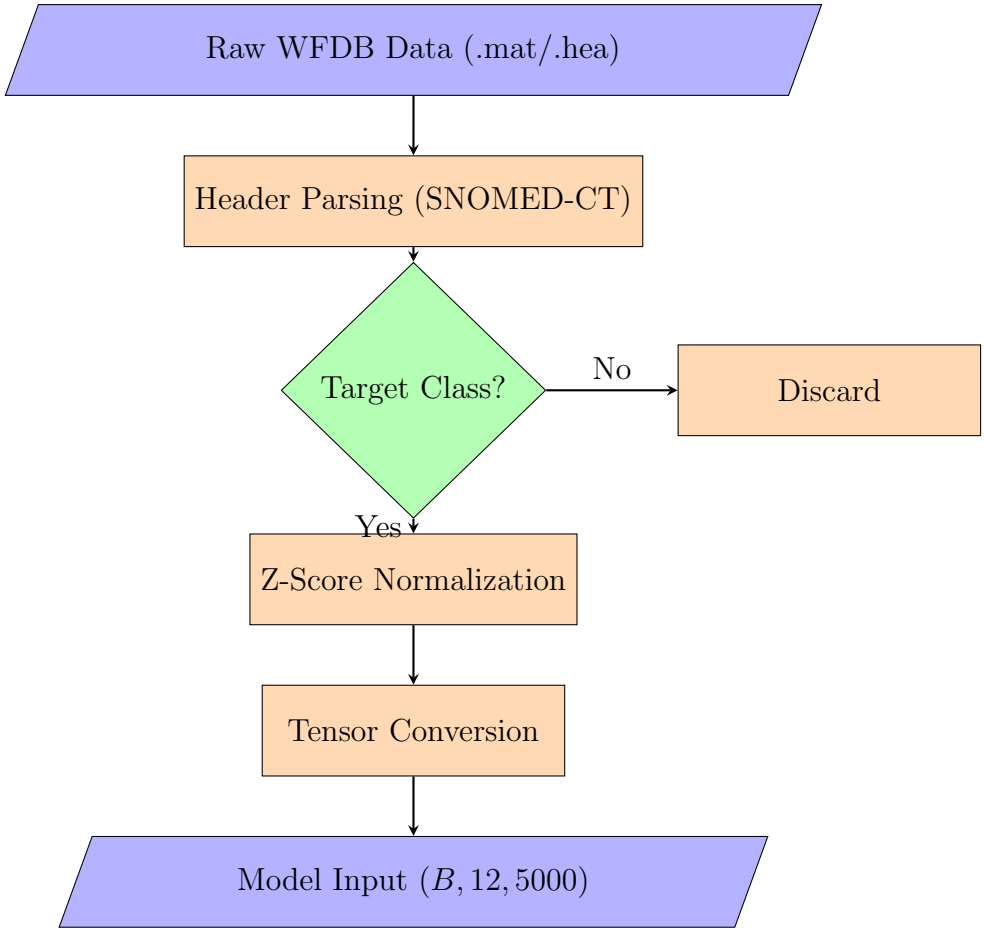

Figure 1: Data Preprocessing Pipeline.

2.2 Preprocessing Steps

1. **Signal Extraction:** Raw signals are loaded from `.mat` files.

2. **Z-Score Normalization:** To handle baseline wander and amplitude variations, each lead is normalized:

$$x_{norm} = \frac{x - \mu}{\sigma + 1e^{-6}} \tag{1}$$

where μ is the mean and σ is the standard deviation of the signal.

3 System Architecture

The model architecture (Figure 2) adopts the hybrid deep learning strategy described in [1], utilizing CNNs for feature extraction and LSTMs for sequence modeling, but introduces ResNet and Attention mechanisms for improved performance on large-scale data.

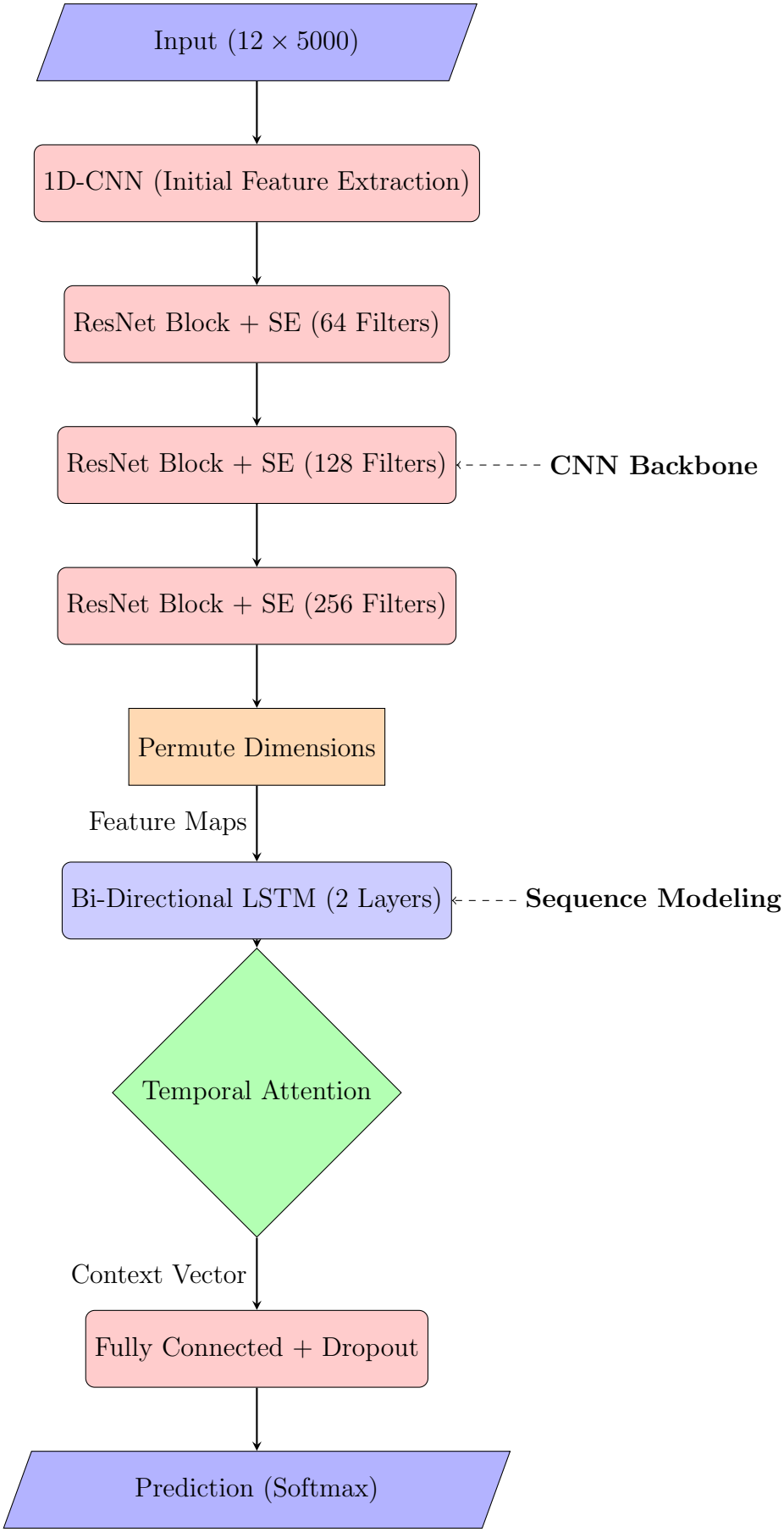

Figure 2: Architecture: SE-ResNet-CNN + BiLSTM.

3.1 Component Details

3.1.1 Convolutional Neural Network (CNN) Backbone

The initial stage is a deep **1D-CNN** that processes the raw 12-lead signal to extract spatial and morphological features (such as the shape of the QRS complex).

- **Residual Blocks (ResNet):** Utilized to prevent vanishing gradients, allowing the model to learn identity mappings and train deeper architectures effectively.
- **Squeeze-and-Excitation (SE) Blocks:** Integrated within the CNN blocks, the SE mechanism adaptively recalibrates channel-wise feature responses, explicitly modeling interdependencies between the 12 ECG leads.

3.1.2 Bidirectional LSTM (BiLSTM)

The feature maps are fed into a 2-layer BiLSTM, capturing long-term temporal dependencies and rhythm patterns in both forward and backward directions.

3.1.3 Temporal Attention

A custom attention layer calculates a weighting score for every time step, allowing the network to focus on critical heartbeats while ignoring noise.

3.2 Implementation Environment

The model was implemented using the **PyTorch** framework. Training was conducted on the **Google Colab** platform utilizing an **NVIDIA T4 GPU**. The training process spanned 35 epochs with a batch size of 64, optimizing for Cross-Entropy Loss weighted to handle class imbalance.

4 Experimental Results

4.1 Quantitative Analysis

Table 1 presents the detailed classification performance for each cardiac condition.

Table 1: Classification Report Metrics

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Sinus Bradycardia (SB)	0.99	0.99	0.99	4718
Sinus Rhythm (SR)	0.98	0.96	0.97	2411
Sinus Tachycardia (ST)	0.96	0.97	0.97	1593
Atrial Flutter (AF)	0.75	0.79	0.77	1692
Atrial Fibrillation (AFIB)	0.27	0.24	0.25	524
Accuracy			**0.91**	10938
Macro Avg	0.79	0.79	0.79	10938
Weighted Avg	0.91	0.91	0.91	10938

4.2 Confusion Matrix Analysis

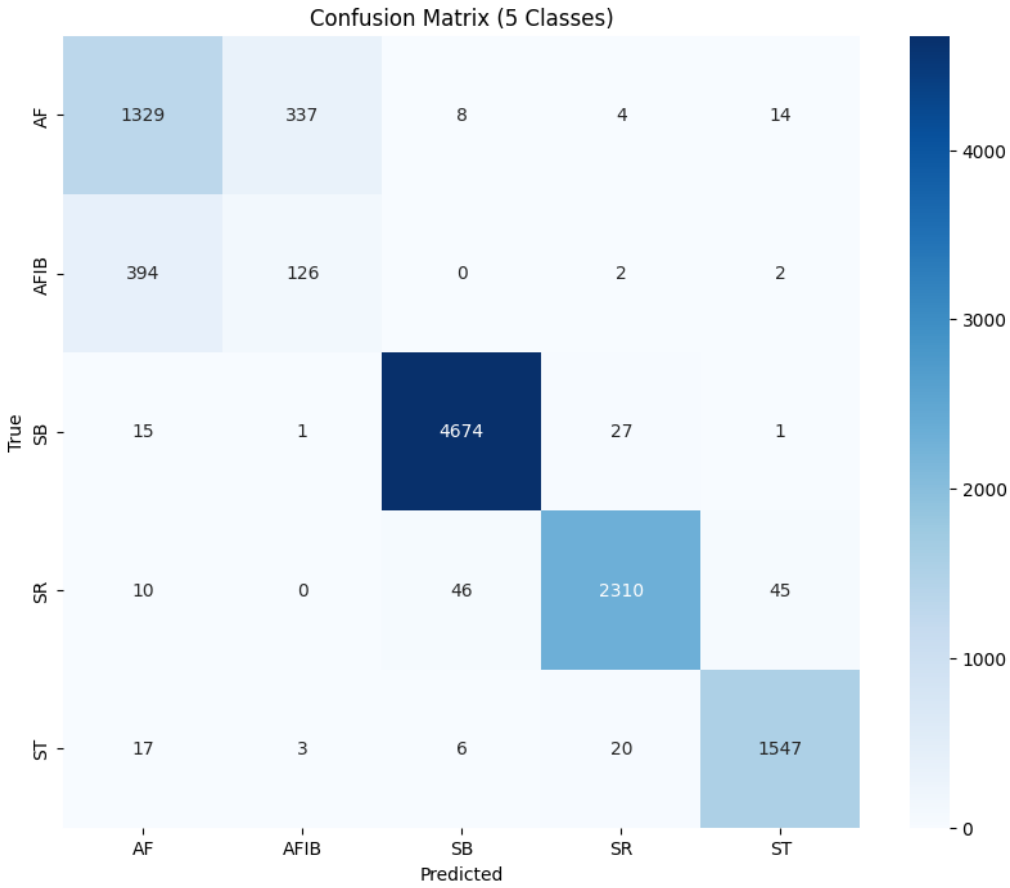

Figure 3: Confusion Matrix (5 Classes).

As shown in Figure 3, the model demonstrates high precision for Sinus Bradycardia (SB) and Sinus Tachycardia (ST). The primary challenge observed is distinguishing Atrial Fibrillation (AFIB) from Atrial Flutter (AF), likely due to the physiological similarities in atrial activity.

4.3 ROC Analysis

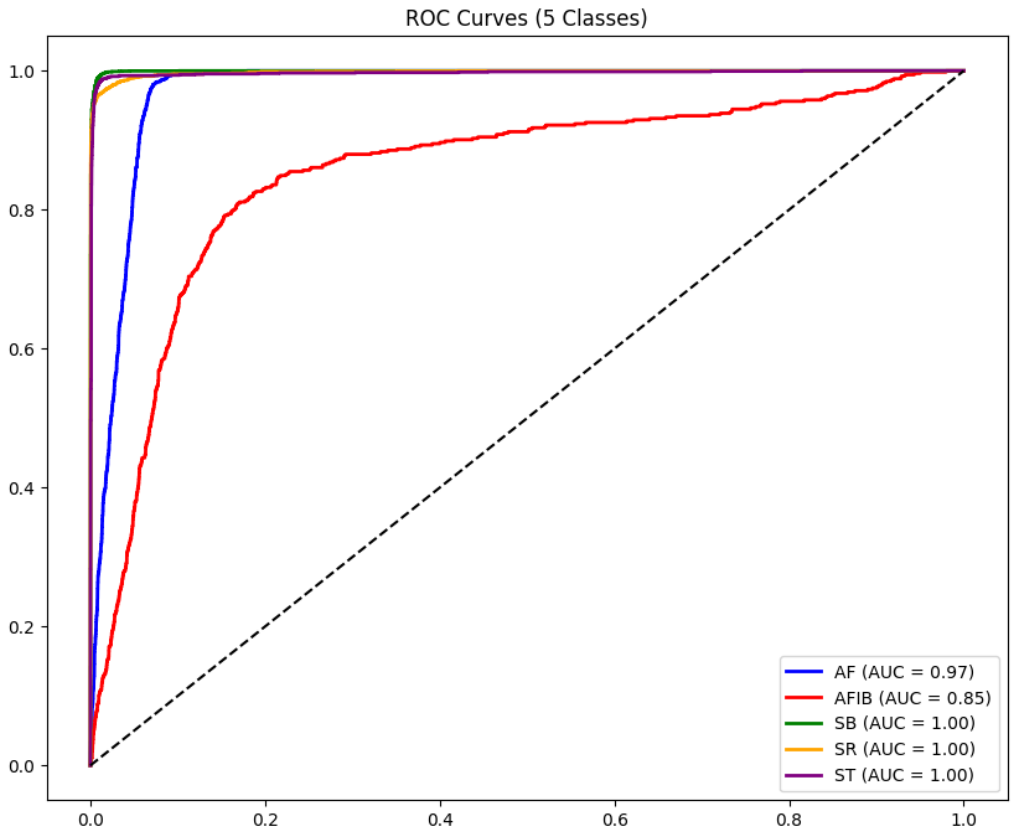

Figure 4: Receiver Operating Characteristic (ROC) Curves.

The ROC curves (Figure 4) confirm the model's robustness, achieving an Area Under Curve (AUC) of 1.00 for sinus rhythm classes and 0.97 for Atrial Flutter.

5 Conclusion

This study successfully implements a hybrid Deep Learning approach for ECG analysis. By combining **ResNet-based CNNs** for robust feature extraction with **BiLSTMs** for temporal analysis—a strategy rooted in the work of Eleyan and Alboghbaish [1]—the system effectively automates the diagnosis of complex arrhythmias. The inclusion of SE blocks and Attention mechanisms further enhances the model's ability to focus on clinically relevant signal components.

References

[1] Eleyan, A., & Alboghbaish, E. (2024). Electrocardiogram Signals Classification Using Deep-Learning-Based Incorporated Convolutional Neural Network and Long Short-Term Memory Framework. *Computers*, 13(2), 55.

[2] Zheng, J., Zhang, J., Danioko, S., Yao, H., Guo, H., & Rakovski, C. (2020). A large-scale 12-lead electrocardiogram database for arrhythmia study. *PhysioNet*.